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APPARATUS AND SYSTEMS FOR ADAPTABLE CASSETTE HVAC BRACKET ASSEMBLY

Abstract

An HVAC cassette unit mounting bracket assembly includes a mounting bracket and a side panel. The mounting bracket includes a first leg and a second leg. The second leg includes one or more inner tabs and second leg holes. The side panel includes one or more first cutouts and first panel holes, one or more second cutouts and second panel holes. A bottom of each cutout is adjacent to a corresponding panel hole. Each first cutout is configured to receive an inner tab. The corresponding first panel hole is configured to be coupled to a second leg hole when the mounting bracket is mechanically coupled to a ceiling joist through ceiling drywall. Each second cutout is configured to receive an inner tab. The corresponding second panel hole is configured to be coupled to a second leg hole when the mounting bracket is mechanically coupled directly to the ceiling joist.

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Background/Summary

TECHNICAL FIELD

[0001] The present embodiments relate to cassette HVAC (heating, ventilation, and air conditioning) units mounted into a ceiling, particularly related to cassette HVAC units mounted to ceiling joists with or without ceiling drywall.

BACKGROUND

[0002] Many buildings use a central HVAC system with cassette units installed in the ceiling or attic to heat or cool specific areas. Installing these units can be a challenge because sometimes (such as in new installations) the units can be installed directly to ceiling joists, and the ceiling drywall is then installed afterward. In such new installations, a particular bracket can be used to install the cassette directly to the ceiling joists. But in other applications (such as existing or retrofit applications), the cassette units are installed after the ceiling drywall is already hung. In these retrofit applications, users must cut holes in the existing ceiling drywall, and, in order to use the same bracket as needed for a new installation, the user must cut the drywall wide enough to expose both ceiling joists. This means that the drywall section now has lost its support from these two adjacent ceiling joists, and the piece of ceiling drywall is therefore lacking support for a span of over 48 inches (for ceiling joists on 16-inch centers), which is not desirable. Accordingly, a need exists for a bracket assembly that can be used for both new installation and retrofit installation, that does not require ceiling drywall anchors to be removed from the two ceiling joists between which the cassette is installed.

SUMMARY

[0003] In some embodiments, a mounting bracket for installation of a ceiling cassette unit, the mounting bracket includes a first leg and a second leg. The first leg is configured to be mechanically coupled to a ceiling joist directly or through ceiling drywall. The second leg includes a first set of second leg holes and a second set of second leg holes. The first set of second leg holes and the second set of second leg holes are configured to be mechanically coupled to panel holes in a side panel attached to the ceiling cassette unit. The first leg and the second leg share a joint edge. The first set of second leg holes is configured to be coupled to the panel holes when the first leg is mechanically coupled to the ceiling joist through the ceiling drywall. The second set of second leg holes is configured to be coupled to the panel holes when the first leg is mechanically coupled directly to the ceiling joist.

[0004] In some embodiments, a mounting bracket assembly for installation of a ceiling cassette unit includes a mounting bracket and a side panel. The mounting bracket includes a first leg and a second leg. The first leg and the second leg share a joint edge. The second leg includes one or more inner tabs. Each inner tab includes one or more second leg holes. The side panel includes one or more first cutouts, one or more first panel holes, one or more second cutouts, and one or more second panel holes. A bottom of each one or more first cutouts is adjacent to a corresponding one or more first panel holes. A bottom of each one or more second cutouts is adjacent to a corresponding one or more second panel holes. Each first cutout is configured to receive one of the inner tabs and the corresponding first panel hole is configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled to a ceiling joist through ceiling drywall. Each second cutout is configured to receive one of the inner tabs and the corresponding second panel holes are configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled directly to the ceiling joist.

[0005] In some embodiments, a ceiling cassette unit assembly, mountable in an opening of a ceiling between a pair of ceiling joists, includes two or more mounting brackets, a housing, and a fan assembly. Each mounting bracket includes a first leg and a second leg. The first leg and the second leg share a joint edge. The second leg includes one or more inner tabs. Each inner tab includes one or more second leg holes. A housing includes first and second side panels, a front panel, and a rear panel, the first and second side panels configured to receive the mounting brackets. Each side panel

includes one or more first cutouts, one or more first panel holes, one or more second cutouts, and one or more second panel holes. A bottom of each one or more first cutouts is adjacent to a corresponding one or more first panel holes. A bottom of each one or more second cutouts is adjacent to a corresponding one or more second panel holes. The fan assembly includes a fan and a motor in the housing. Each first cutout is configured to receive one of the inner tabs and the corresponding first panel hole is configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled to the ceiling joist through ceiling drywall. Each second cutout is configured to receive one of the inner tabs and the corresponding second panel holes are configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled directly to the ceiling joist.

[0006] These and additional features provided by the embodiments of the present disclosure will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the disclosure. The following detailed description of the illustrative embodiments can be better understood when read in conjunction with the following drawings, wherein like structure is indicated with like reference numerals and in which:

[0008] FIG. 1 is a schematic view of a mounted HVAC cassette unit using a mounting bracket assembly, according to one or more embodiments shown and described herein;

[0009] FIG. 2A is a section view of an example mounting bracket assembly with a cassette unit installation shown mounted to ceiling joists through ceiling drywall, according to one or more embodiments shown and described herein;

[0010] FIG. 2B is a section view of the example mounting bracket assembly and cassette unit of FIG. 2A, shown mounted directly to ceiling joists, according to one or more embodiments shown and described herein;

[0011] FIG. 3 is a schematic view of an example mounting bracket assembly for cassette unit installation, according to one or more embodiments shown and described herein;

[0012] FIG. 4A is a front view of a portion of an example mounting bracket assembly for cassette unit installation, shown disengaged from a side panel, according to one or more embodiments shown and described herein;

[0013] FIG. 4B is a front view of a portion of an example mounting bracket assembly for cassette unit installation, shown engaged with a side panel, according to one or more embodiments shown and described herein;

[0014] FIG. 4C is a section view of a portion of an example mounting bracket assembly for cassette unit installation, shown disengaged from a side panel, according to one or more embodiments shown and described herein;

[0015] FIG. 4D is a section view of a portion of an example mounting bracket assembly for cassette unit installation, shown engaged with a side panel, according to one or more embodiments shown and described herein; and

[0016] FIG. 5 is a schematic view of an example mounting bracket assembly for cassette unit installation, in accordance with one or more embodiments shown and described herewith.

DETAILED DESCRIPTION

[0017] The embodiments described herein are directed to apparatus and systems for mounting bracket assemblies and ceiling cassette unit systems. The disclosed mounting bracket assembly may include a mounting bracket and a corresponding side panel of the cassette. The mounting

bracket may include a first leg and a second leg. The first leg and the second leg may share a joint edge. The second leg may include one or more inner tabs. Each inner tab may include one or more second leg holes. The side panel may include one or more first cutouts and one or more first panel holes, and one or more second cutouts and one or more second panel holes. A bottom of each said one or more first cutouts may be adjacent to a corresponding one or more first panel holes. A bottom of each said one or more second cutouts may be adjacent to a corresponding one or more second panel holes. Each first cutout may be configured to receive one of the inner tabs and the corresponding first panel hole may be configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled to ceiling drywall. Each second cutout may be configured to receive one of the inner tabs and the corresponding second panel holes may be configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled to a ceiling joist directly.

[0018] Users installing cassette units using current mounting brackets encounter challenges when faced with differing types of installations. Standard brackets cannot adequately handle all types of installations. The bracket used for direct installation to the joists is not suitable for installing over drywall. Similarly, the bracket used for installing over drywall is not suitable for directly installing to the joists. This limitation necessitates multiple brackets, which increases cost, or requires project adjustments, potentially compromising the design or mandating additional support structures.

[0019] In contrast, the disclosed mounting bracket assemblies provide a solution for installing ceiling cassette units on joists in both types of installation scenarios: (a) directly to the ceiling joists in the absence of ceiling drywall (as seen in new construction) and (b) to the ceiling joists through existing ceiling drywall. This design streamlines the installation process, eliminating the need for a separate bracket, additional materials, and removing too much drywall. This adaptability not only simplifies installation but also enhances efficiency, reducing the reliance on various bracket designs or intricate adjustments. The mounting bracket assemblies offer flexible options for installation, enabling both new installations and retro-fit installations using a single bracket. As such, using the mounting brackets described herein can reduce installation time and prevent the need for removing the drywall anchors that are already inserted into the ceiling joists.

[0020] The embodiments discussed hereinafter will focus on the implementation of the hereinafter-described techniques and apparatus within a residential, commercial, and/or building structure, such as the type that may be used in single-family or multi-family dwellings, or in other similar applications. However, it will be appreciated that the herein-described techniques and apparatus may also be used in connection with other types of building structures in some embodiments. For example, the herein-described techniques may be used in commercial applications in some embodiments. Moreover, for example, the cassette HVAC units mounted in a ceiling may be a four-way cassette or two-way cassette in some embodiments. Similarly, the particulars of the HVAC cassette **201** will not be described in detail, as this disclosure relates to the mounting of such units, rather than the internal operation of such units. Therefore, many features of such units will not be described in detail (such as, for example, a fan, a fan motor, evaporator coils, condensate pump, controls, conduits, and the like), it being understood that the teachings of this disclosure apply to many types of HVAC cassette units.

[0021] Referring now to figures, FIG. **1** generally depicts an HVAC cassette unit **201** installed in a spacing **417** between two parallel ceiling joists **411**. FIG. **2A** depicts a retro-fit installation, where the HVAC cassette unit **201** is fastened to the ceiling joists **411** through drywall **413**. FIG. **2B** depicts a new installation, where the HVAC cassette unit **201** is fastened to the ceiling joists **411** directly.

[0022] The HVAC cassette unit **201** includes a housing **203** to hold different components of the HVAC cassette unit **201**. The housing **203** may include side panels **107** (e.g., a first side panel **107a** and a second side panel **107b**), a front panel **109**, and a rear panel. The side panels **107** work in conjunction with a mounting bracket (described below) to enable the HVAC cassette unit **201** to be

directly mounted on the ceiling joists **411** (e.g., as in FIG. 2B) or indirectly through the ceiling drywall **413** (e.g., as illustrated in FIG. 2A) through mechanical coupling with one or more of mounting brackets **101**.

[0023] In typical installations, the ceiling joists **411** are spaced from each other to define a spacing **417** therebetween. The spacing **417** between two parallel ceiling joists **411** allows a HVAC cassette unit **201** to be secured to the ceiling joists **411** within the spacing **417**. Spacing **417** can be any spacing, such as common spacings like on-center spans of 12 inches, 16 inches, 24 inches, and the like. In new installations of HVAC cassette unit **201**, often there is no drywall **413** present, so the HVAC cassette unit **201** is installed directly to the ceiling joists **411**, as will be described below and as shown in FIG. 2B. In retro-fit installations, ceiling drywall **413** may be present, such that a ceiling drywall opening **418** must be created in order to install the HVAC cassette unit **201**. The mounting bracket assembly **100** described herein enables a smaller ceiling drywall opening **418** to be used to mount the HVAC cassette unit **201**, as compared to prior art brackets (which require the ceiling drywall opening **418** to expose not only the ceiling joist spacing **417**, but also the width of both ceiling joists **411**). As will be described, the mounting bracket assembly **100** advantageously enables a user to cut the drywall opening **418** to match the ceiling joint spacing **417**, thus allowing the ceiling drywall fasteners beneath the ceiling joists **411** to remain in place.

[0024] FIG. 3 illustrates a mounting bracket **101** that enables both new and retro-fit installations. The mounting bracket **101** includes a first leg **105** and a second leg **103**. The first leg **105** and the second leg **103** may share a joint edge **135**. The first leg **105** may be configured to be mechanically coupled to a ceiling joist **411** either directly or through the ceiling drywall **413**. The second leg **103** may be configured to be mechanically coupled to the side panel **107** of the HVAC cassette unit **201**. The first leg **105** may include attachment mechanisms to allow the mounting bracket **101** to be configured to engage with the ceiling joist **411** directly or through the ceiling drywall **413**. For example, the first leg **105** may include a set of first leg holes **151**. A first set of fasteners **251**, for example, can be inserted through the first leg holes **151** and into the ceiling joists **411** (directly (FIG. 2B) or through the ceiling drywall **413** (FIG. 2A)).

[0025] In some embodiments, the second leg **103** may include a set of inner tabs **131** extending from near the joint edge **135** away from the first leg **105** in the z-direction as shown in FIG. 3. The inner tabs **131** may be at least partially circumscribed within a surface **132** of the second leg **103**. Two or more extended edges **181**, **182** of each inner tab **131** have a gap **134** between the surface **132** and each respective extended edge **181**, **182**, such that the gap **134** between the inner tabs **131** and the surface **132** is in an upside-down U-shape (e.g., as illustrated in FIGS. 3-4B). Alternatively, the inner tab **131** may be triangular in shape, such that the two extended edges **181**, **182** are angled towards each other, in order to create an upside-down V-shape gap **134** between the surface **132** and each extended edge **181**, **182**, respectively). The inner tabs **131** have a bottom portion **140** that is configured to mechanically couple with various cutouts **175**, **177** in the side panel **107** (described below) to mechanically couple the HVAC cassette unit **201** to the mounting bracket **101**.

[0026] With reference to FIGS. 3-4D, each inner tab **131** may also include one or more second leg hole **133** (e.g., as illustrated in FIG. 4A-4D). The second leg holes **133** may be configured to be aligned with a set of panel holes **171** and **173** of the side panel **107**. In some embodiments, a second set of fasteners **261** may be inserted through the second leg holes **133** and into the panel holes **171** and **173** such that the second leg **103** is additionally mechanically attached to the side panel **107** via second set of fasteners **261**. The panel holes **171** and **173** may include bosses. For example, the panel holes **171** and **173** may include threaded bosses and the second set of fasteners **261** may be threadably coupled to the threaded bosses.

[0027] As discussed, the side panel **107** may include a set of first cutouts **177**, a set of first panel holes **173**, a set of second cutouts **175**, and a set of second panel holes **171**. The set of first cutouts **177** may have a bottom **190**. The bottom **190** is located at a given depth **191** from a lower edge of the side panel **107**. The set of second cutouts may have a bottom **195**. The bottom **195** is located at

a given depth **196** from the lower edge of the side panel **107**. The set of first panel holes **173** is located adjacent the bottom **190**. The set of second panel holes **171** is located adjacent the bottom **195**. For example, as illustrated in FIG. 3-4D, each set of first panel holes **173** may be positioned above the bottom of a corresponding first cutout **177** in the z-direction of the coordinate, and each set of second panel holes **171** may be positioned above the bottom of a corresponding second cutout **175** in the z-direction of the coordinate. The distances between the bottoms **190**, **195** of the cutouts **177**, **175**, respectively, and their corresponding panel holes **173**, **171** may be approximately equal to or less than distances between the second leg holes **133** to the joint edge **135**.

[0028] In some embodiments, as illustrated in FIG. 3, the depth **191** of the first cutout **177** may be less than the depth **196** of the second cutout **175**. The depth difference between the first cutouts **177** and the second cutouts **175** may be approximately equal to a thickness of the ceiling drywall **413**. The first cutout **177** may be adjacent to the second cutout **175** in the y-direction of the coordinate such that a left edge of the first cutout **177** is shared as a right edge of the second cutout **175**. As such, a neighboring first cutout **177** and a neighboring second cutout **175** may merge into a continuous cutout with shared edge.

[0029] With continued reference to FIGS. 2A-4D, the inner tabs **131** and the second leg holes **133** may be configured to be mechanically coupled to the first cutouts **177** and the first panel holes **173** of the side panel **107**, respectively, when the first leg **105** is mechanically coupled through the ceiling drywall **413**. Alternatively, the inner tabs **131** and the second leg holes **133** may be configured to be mechanically coupled to the second cutouts **175** and the second panel holes **171** of the side panel **107**, respectively, when the first leg **105** is mechanically coupled directly to the ceiling joists **411**. For example, as illustrated in FIGS. 3-4D, each first cutout **177** is configured to receive one of the inner tabs **131**, and the corresponding first panel hole **173** is configured to be coupled to one of the second leg holes **133** within the corresponding inner tab **131** when the mounting bracket **101** is mechanically coupled to the ceiling joists **411** through the ceiling drywall **413** (e.g., as illustrated in FIG. 2B). Alternatively, as illustrated in FIGS. 3-4D, each second cutout **175** is configured to receive one of the inner tabs **131** and the corresponding second panel hole **171** is configured to be coupled to one of the second leg holes **133** within the corresponding inner tab **131** when the mounting bracket **101** is mechanically coupled directly to the ceiling joists **411** (e.g., as illustrated in FIG. 2A). The cutouts **175** and **177** may be configured to receive the inner tabs **131** to allow the inner tabs **131** to insert behind the side panel **107** such that an inner surface **573** of the side panel **107** mechanically contacting with the inner tabs **131** and the rest part of the second leg **103** (i.e., the surface **132** of the second leg **103**) contact with the outer surface **571** of the side panel **107**. Accordingly, the inner tabs **131** and the surface **132** of the second leg **103** may sandwich the side panel **107** in between in a side view of the mounting bracket assembly **300** (e.g., as illustrated in FIG. 4D). In some embodiments, the inner tabs **131** may be inserted behind the side panel **107** until the bottom **190**, **195** of the cutouts **177** and **175** are mechanically coupled to a portion of the inner tabs **131**, such as the bottom portion **140** of the inner tabs **131**. In some embodiments, the inner tabs **131** may be inserted behind the side panel **107** to the extent of allowing the first panel holes **173** or the second panel holes **171** to match with the second leg holes **133**, respectively.

[0030] As discussed, the mounting bracket assembly **100** enables a user to mount the HVAC cassette unit **201** within the opening **418** directly to the ceiling joists **411** before installing the ceiling drywall **413** or, alternatively, to ceiling joists **411** after ceiling drywall **413** has been installed, by cutting an opening **418** in the drywall **413** that is approximately the same width as the spacing **417**. In either scenario, the first side panel **107a** and the second side panel **107b** are configured to be engaged with the inner tabs **131** in the inner surface **573** such that either the bottom **190** of the first cutouts **177** or the bottom **195** of the second cutouts **175** are mechanically coupled to a bottom portion **140** of the inner tabs **131**. The design of the mounting bracket assembly **100** allows the mounting bracket **101**, without being fully disengaged from the side panel **107**, to have its second leg **103** selectively moved to couple to either the first cutouts **177** or the

second cutouts **175**. For example, a user may continuously move and slide the mounting bracket **101** within the in-plane directions (e.g., the y-direction in the coordinate of FIGS. 3-5B) with the inner tabs **131** behind the side panel **107** (e.g., the inner tabs **131** continuously contacting the inner surface **573** of the side panel **107**).

[0031] In retro-fit installations (FIG. 2A), the inner tab **131** of the mounting brackets **101** may be inserted behind the first side panel **107a** and the second side panel **107b** to contact with the inner surface **573** of the side panels **107** such that the second leg hole **133** of the mounting bracket **101** matches the first panel hole **173** to allow the second set of fasteners **261** to secure the mounting bracket **101** relative to the HVAC cassette unit **201**. Further, the first leg hole **151** can be configured to be mechanically coupled to the ceiling joists **411** through the ceiling drywall **413** to secure the mounting bracket **101** related to the ceiling joists **411**. Accordingly, the HVAC cassette unit **201** is mounted to the ceiling joists **411**.

[0032] Alternatively, in new installations (FIG. 2B), the inner tab **131** of the mounting brackets **101** may be inserted behind the first side panel **107a** and the second side panel **107b** to contact with the inner surface **573** of the side panels **107** such that the second leg hole **133** of the mounting bracket **101** matches the second panel hole **171** to allow the second set of fasteners **261** to secure the mounting bracket **101** relative to the HVAC cassette unit **201**. Further, the first leg hole **151** is configured to be mechanically coupled to the ceiling joists **411** through the ceiling drywall **413** to secure the mounting bracket **101** related to the ceiling joists **411**. Accordingly, the HVAC cassette unit **201** is mounted to the ceiling joists **411**. The ceiling drywall **413** with the opening **418** may be then installed to have the bottom of the ceiling joists **411** covered such that the ceiling drywall **413** may be screwed to the ceiling joists **411**.

[0033] It should be appreciated that in alternative embodiments, some of the mechanical structures of the side panel **107** and the mounting bracket **101** may be exchangeable. For example, the second leg **103** may include cutouts while the side panel **107** may include inner tabs, in a manner similar to that described above.

[0034] Referring now to FIG. 5, an alternative embodiment of a mounting bracket assembly **300** for installation of the HVAC cassette unit **201** is depicted. The mounting bracket **101** may include a first leg **105** and a second leg **103**. The first leg **105** and the second leg **103** may share a joint edge **135**. The first leg **105** may be configured to be mechanically coupled to a ceiling joist **411** either directly or through the ceiling drywall **413**. The second leg **103** may be configured to be mechanically coupled to the side panel **107**. The second leg **103** may include a first set of second leg holes **531** and a second set of second leg holes **533**.

[0035] The first set of second leg holes **531** may be above the second set of second leg holes **533** in the z-direction as shown in FIG. 5. The distance between the first set of second leg holes **531** and the second set of second leg holes **533** may be essentially equal to or more than a thickness of ceiling drywall **413**. In some embodiments, a first distance between the first set of second leg holes **531** and the joint edge **135** is less than a second distance between the second set of second leg holes **533** and the joint edge **135**. The difference between the first distance and the second distance is essentially equal to or more than a thickness of the ceiling drywall **413**. The first leg **105** may include attachment mechanisms, such as a set of first leg holes **151**. The attachment mechanisms such as the first leg holes **151** are configured to attach the first leg **105** to the ceiling joists **411** either directly or through the ceiling drywall **413**. The side panel **107** of the mounting bracket assembly **300** may include a set of panel holes **171**. The first set of second leg holes **531** and the second set of second leg holes **533** may be configured to be mechanically coupled to the set of panel holes **171** of the side panel **107**. In some embodiments, a second set of fasteners **261** may be inserted through the first set of second leg holes **531** or the second set of second leg holes **533** to further mate with the set of panel holes **171** such that the second leg **103** is mechanically attached to the side panel **107**. The set of panel holes **171** may include bosses. For example, the panel holes **171** may include threaded bosses and the second set of fasteners **261** may be threadably coupled to

the threaded bosses.

[0036] Similar to FIGS. 2A-2B, a user may decide to mount the HVAC cassette unit **201** to the ceiling joists **411** directly or through the ceiling drywall **413** through the mounting bracket assembly **300**. For example, the HVAC cassette unit **201** may be mounted or secured through the ceiling drywall **413** by inserting first set of fasteners **251** through the set of first leg holes **151** and into the ceiling joists **411**. The mounting bracket **101** is mounted to the HVAC cassette unit **201** by inserting second set of fasteners **261** through the first set of second leg holes **531** and through the set of panel holes **171**. Alternatively, the HVAC cassette unit **201** may be mounted or secured to the ceiling joists **411** directly by inserting first set of fasteners **251** through the set of first leg holes **151** and into the ceiling joists **411**. The mounting bracket **101** is mounted to the HVAC cassette unit **201** by inserting second set of fasteners **261** through the second set of second leg holes **533** and through the set of panel holes **171**.

[0037] In some embodiments, a method of installation of the HVAC cassette unit **201** including the mounting bracket **101** and the side panels **107** may include determining whether ceiling drywall **413** is installed. In new installations, or where ceiling drywall **413** is not installed, the method includes inserting the inner tabs **131** of the second leg **103** to the second cutouts **175** to mechanically couple the second leg holes **133** with the second panel holes **171**. The method may include arranging the HVAC cassette unit **201** between a pair of ceiling joists **411**. The method may include mechanically coupling the first leg **105** directly to the ceiling joists **411**.

[0038] In some embodiments, the method of installation of the HVAC cassette unit **201** includes determining the ceiling drywall **413** is already installed. In this case a ceiling drywall opening **418** is created to open the ceiling drywall to a width that approximately matches spacing **417**. The method includes inserting the inner tabs **131** of the second leg **103** to the first cutouts **177** to mechanically couple the second leg holes **133** with the first panel holes **173**. The method may include arranging the HVAC cassette unit **201** between a pair of ceiling joists **411** through the opening **418**. The method may include mechanically coupling the first leg **105** to the ceiling joists **411** through the ceiling drywall **413**.

[0039] While several embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, embodiments may be practiced otherwise than as specifically described and claimed. Embodiments of the present disclosure are directed to each individual feature, system, article, material, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, and/or methods, if such features, systems, articles, materials, and/or methods are not mutually inconsistent, is included within the scope of the present disclosure.

[0040] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined. The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

[0041] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are

conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0042] As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0043] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0044] It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

[0045] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively.

[0046] It is to be understood that the embodiments are not limited in their application to the details of construction and the arrangement of components set forth in the description or illustrated in the drawings. The invention is capable of some embodiments and of being practiced or of being carried out in various ways. Unless limited otherwise, the terms “connected,” “coupled,” “in communication with,” and “mounted,” and variations thereof herein are used broadly and encompass direct and indirect connections, couplings, and mountings. In addition, the terms “connected” and “coupled” and variations thereof are not restricted to physical or mechanical connections or couplings.

[0047] The foregoing description of several embodiments of the invention has been presented for purposes of illustration. It is not intended to be exhaustive or to limit the invention to the precise

steps and/or forms disclosed, and obviously many modifications and variations are possible in light of the above teaching.

Claims

- 1.** A mounting bracket for installation of a ceiling cassette unit, the mounting bracket comprising: a first leg configured to be mechanically coupled to a ceiling joist directly or through ceiling drywall; a second leg comprising a first set of second leg holes and a second set of second leg holes, said first set of second leg holes and said second set of second leg holes configured to be mechanically coupled to panel holes in a side panel attached to the ceiling cassette unit, wherein: the first leg and the second leg share a joint edge; the first set of second leg holes is configured to be coupled to the panel holes when the first leg is mechanically coupled to the ceiling joist through the ceiling drywall; and the second set of second leg holes is configured to be coupled to the panel holes when the first leg is mechanically coupled directly to the ceiling joist.
- 2.** The mounting bracket of claim 1, wherein the second leg further comprises one or more inner tabs extending from near the joint edge between the first leg and the second away from the first leg.
- 3.** The mounting bracket of claim 2, wherein one or more holes of the first set of second leg holes or of the second set of the second leg holes is positioned in each inner tab.
- 4.** The mounting bracket of claim 2, wherein the inner tabs are configured to contact an inner surface of the side panel when a plate of the second leg contacts an outer surface of the side panel.
- 5.** The mounting bracket of claim 2, wherein the inner tabs are substantially within a plate structure of the second leg, wherein two or more extended edges of each inner tab have a gap between the plate structure and the inner tab.
- 6.** The mounting bracket of claim 1, wherein the first leg comprises a set of first leg holes configured to engage with a fastener to mechanically couple the first leg to the ceiling joist or the ceiling drywall.
- 7.** The mounting bracket of claim 1, wherein the first leg is mechanically coupled to the ceiling joist through the ceiling drywall such that the ceiling drywall is mechanically coupled to the ceiling joist.
- 8.** The mounting bracket of claim 1, wherein a first distance between the first set of second leg holes and the joint edge is less than a second distance between the second set of second leg holes and the joint edge.
- 9.** The mounting bracket of claim 8, wherein a difference between the first distance and the second distance is essentially equal to or more than a thickness of the ceiling drywall.
- 10.** A mounting bracket assembly for installation of a ceiling cassette unit comprising: a mounting bracket comprising a first leg and a second leg, wherein: the first leg and the second leg share a joint edge; the second leg comprises one or more inner tabs; and each inner tab comprises one or more second leg holes; a side panel comprising: one or more first cutouts and one or more first panel holes, wherein a bottom of each said one or more first cutouts is adjacent to a corresponding one or more first panel holes; one or more second cutouts and one or more second panel holes, wherein a bottom of each said one or more second cutouts is adjacent to a corresponding one or more second panel holes, wherein: each said first cutout is configured to receive one of the inner tabs and the corresponding first panel hole is configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled to a ceiling joist through ceiling drywall; and each said second cutout is configured to receive one of the inner tabs and the corresponding second panel holes are configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled directly to the ceiling joist.
- 11.** The mounting bracket assembly of claim 10, wherein an inner surface of the side panel is configured to contact the inner tab, and an outer surface of the side panel is configured to contact a plate structure of the mounting bracket.

- 12.** The mounting bracket assembly of claim 10, wherein the bottom of each first cutout and the bottom of each second cutout are configured to be mechanically coupled to a portion of one of the inner tabs.
- 13.** The mounting bracket assembly of claim 10, wherein the first cutouts have a depth less than the second cutouts.
- 14.** The mounting bracket assembly of claim 13, wherein a depth difference between the first cutouts and the second cutouts is approximately equal to a thickness of the ceiling drywall.
- 15.** A ceiling cassette unit assembly, mountable in an opening of a ceiling between a pair of ceiling joists, comprising: two or more mounting brackets, each mounting bracket comprising a first leg and a second leg, wherein: the first leg and the second leg share a joint edge; the second leg comprises one or more inner tabs; and each inner tab comprises one or more second leg holes; a housing comprising first and second side panels, a front panel, and a rear panel, the first and second side panels configured to receive the mounting brackets, each side panel comprising: one or more first cutouts and one or more first panel holes, wherein a bottom of each said one or more first cutouts is adjacent to a corresponding one or more first panel holes; one or more second cutouts and one or more second panel holes, wherein a bottom of each said one or more second cutouts is adjacent to a corresponding one or more second panel holes; a fan assembly comprising a fan and a motor in the housing; and wherein: each said first cutout is configured to receive one of the inner tabs and the corresponding first panel hole is configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled to the ceiling joist through ceiling drywall; and each said second cutout is configured to receive one of the inner tabs and the corresponding second panel holes are configured to be coupled to one of the second leg holes when the mounting bracket is mechanically coupled directly to the ceiling joist.
- 16.** The ceiling cassette unit assembly of claim 15, wherein the first leg of each mounting bracket is configured to be mechanically coupled to one of the ceiling joists before the ceiling drywall is installed.
- 17.** The ceiling cassette unit assembly of claim 15, wherein the inner tabs are substantially within a plate structure of the second leg, wherein two or more extended edges of each inner tab has a gap between the plate structure, wherein the gap is a U-shape gap or a V-shape gap.
- 18.** The ceiling cassette unit assembly of claim 15, wherein the side panel is configured to be engaged with the inner tab in an inner surface of the side panel.
- 19.** The ceiling cassette unit assembly of claim 15, wherein the bottom of each first cutout and the bottom of each second cutout are configured to be mechanically coupled to a bottom portion of the inner tab.
- 20.** The ceiling cassette unit assembly of claim 15, wherein the first cutouts have a depth less than the second cutouts, and a depth difference between the first cutouts and the second cutouts is essentially equal to a thickness of the ceiling drywall.
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